

Random Forest Vs Bagging

Key Differences

1) Data Sampling

Bagging

Randomly samples data with replacement (Bootstrapping)

Random Forest

Same as bagging: uses bootstrapping for data sampling

2) Feature Selection

Considers all features for every split in every tree

Considers only a random subset of feature at each split

3) Base Model

Can use any algorithm (eg. SVM, KNN, Decision Tree)

Specifically uses Decision Tree.

4) Goal

Primarily reduces variance

Reduces variance & correlation between trees

5) Complexity

Generally simple, trees can be very

More complex due to the additional

similar if one

Feature is dominant

Feature

randomization.